Investment Analysis High-Difficulty Quiz

Multiple Choice · Chapters 3-7 · 11 questions

1. You sell short 100 shares at $50 per share and post the initial margin of 50% in cash. The maintenance margin is 30%. At what share price will you receive a margin call?
   a. $53.85
   b. $57.69
   c. $62.50
   d. $65.00

2. The stock market returns 8% a year before costs. You invest $10,000 for 20 years in either an index fund with a 0.05% expense ratio or an active fund with a 1.00% expense ratio that earns the market return before fees. How much more does the index fund leave you with? (Net return = gross return − expense ratio.)
   a. $9,580
   b. $7,483
   c. $3,720
   d. $1,900

3. You invest $10,000 for one year at a gross portfolio return of 10%. Class A has a 5% front-end load and 0.90% annual expenses. Class C has no front-end load, 1.60% annual expenses, and a 1% CDSC charged on the redemption value if you redeem in year 1. (Net growth = gross return − expense ratio.) What are the ending values after all charges?
   a. Class A $10,364.50 and Class C $10,840.00
   b. Class A $10,910.00 and Class C $10,731.60
   c. Class A $10,450.00 and Class C $10,840.00
   d. Class A $10,364.50 and Class C $10,731.60

4. Class A has a 5% front-end load and 0.90% annual expenses. Class C has no load and 1.60% annual expenses (ignore any CDSC). The gross return is 10% a year. After roughly how many years does Class A overtake Class C?
   a. About 3 years
   b. About 5 years
   c. About 8 years
   d. About 12 years

Exhibit A · Share trades and dividends

Used in questions 5–6

Trades and dividends (KRW per share)

Time	Action	Price	Dividend
t = 0	Buy 1 share	40,000	—
t = 1	Buy 1 more share	44,000	1,000
t = 2	Sell 2 shares	46,000	1,000

5. [Exhibit A] Using the share trades in Exhibit A, what is your dollar-weighted return (IRR) per year? Dividends are paid per share held just before each trade.

a. 9.62%
b. 9.66%
c. 6.82%
d. 8.70%

6. [Exhibit A] Using the share trades in Exhibit A, what is your time-weighted (geometric) return per year? Dividends are paid per share held just before each trade.

a. 9.62%
b. 9.66%
c. 8.70%
d. 12.50%

7. A preferred stock (par value $100) pays an 8% annual dividend rate, paid quarterly. The first dividend will be paid 3 years from today (at the end of quarter 12). The required return is 10% per year, compounded monthly. What is the price today?

a. $58.85
b. $60.33
c. $60.97
d. $79.34

8. Mirae Logistics paid a dividend of $1.35 per share four years ago and has just paid $1.92 per share. Assume dividends keep growing forever at the same average annual rate as over the past four years. The required return is 13%. What is the price today?

a. $36.13
b. $50.59
c. $55.25
d. $86.84

9. A stock pays no dividend in years 1 and 2 and pays $1.80 at the end of year 3. Dividends then grow 20% per year in years 4, 5 and 6, and 5% per year forever after year 6. The required return is 14%. What is the price today?

a. $21.79
b. $21.00
c. $24.84
d. $19.76

10. Gyeongbok Software has just paid a dividend of $2.50 per share. Dividends will grow 18% per year for the next 6 years and 5% per year forever after that. The required return is 14%. Use the two-stage growth model. What is the price today?

a. $32.78
b. $35.87
c. $51.12
d. $52.83

11. Namsan Electronics has just paid a dividend of $1.00 per share and reported EPS of $3.00 for the year just ended. Dividends grow 12% per year for the next 5 years, and EPS grows 10% per year. At the end of year 5 the stock is sold at a forward P/E of 14 (price divided by the next year's EPS). The required return is 12%. What is the price today?

a. $42.22
b. $42.70
c. $43.38
d. $47.22

Investment Analysis High-Difficulty Quiz — Answers

1. b
Assets = 5,000 + 2,500 = $7,500 (sale proceeds + 50% margin)
Liability = 100P (value of the shares owed)
Margin = (7,500 − 100P) / 100P = 0.30
7,500 = 130P, so P = 7,500 / 130 = $57.69
BA II Plus: 7500 ÷ 130 = 57.69

2. b
Index net return = 8% − 0.05% = 7.95%
Active net return = 8% − 1.00% = 7.00%
Index FV = $10{,}000 \times (1.0795)^{20} = \$46{,}180$
Active FV = $10{,}000 \times (1.07)^{20} = \$38{,}697$
Difference = 46,180 − 38,697 = $7,483
BA II Plus: 20 N, 7.95 I/Y, −10000 PV, 0 PMT, CPT FV = 46,180
BA II Plus: 20 N, 7 I/Y, −10000 PV, 0 PMT, CPT FV = 38,697

3. d
Class A growth = 10% − 0.90% = 9.10%
Class C growth = 10% − 1.60% = 8.40%
Class A = $10{,}000 \times 0.95 \times 1.091 = \$10{,}364.50$ (5% load taken at purchase)
Class C = $10{,}000 \times 1.084 \times 0.99 = \$10{,}731.60$ (1% CDSC on redemption)
Class C is ahead after one year
BA II Plus: 9500 × 1.091 = 10,364.50
BA II Plus: 10840 × 0.99 = 10,731.60

4. c
Set $0.95 \times (1.091)^n = (1.084)^n$
$n = \ln(1 / 0.95) / \ln(1.091 / 1.084)$
$\ln(1 / 0.95) = 0.05129$
$\ln(1.091 / 1.084) = 0.006437$
$n = 0.05129 / 0.006437 = 7.97$ years
Check at n = 8: $0.95 \times (1.091)^8 = 1.9069 > (1.084)^8 = 1.9065$
BA II Plus: 1 ÷ 0.95 = LN → 0.05129
BA II Plus: 1.091 ÷ 1.084 = LN → 0.006437
then 0.05129 ÷ 0.006437 = 7.97

5. d [Exhibit A]
$CF_0 = -40{,}000$ (buy 1 share)
$CF_1 = -44{,}000 + 1{,}000 = -43{,}000$ (buy a second share, receive the dividend on 1 share)
$CF_2 = 2 \times 1{,}000 + 2 \times 46{,}000 = +94{,}000$ (dividends on 2 shares, sell 2 shares)
Equation: $0 = -40{,}000 - 43{,}000 / (1 + r) + 94{,}000 / (1 + r)^2$
r = 8.70%
BA II Plus: CF, CF0 = −40000, C01 = −43000, F01 = 1, C02 = 94000, F02 = 1, IRR CPT = 8.70

6. a [Exhibit A]
Year 1 return = (44,000 − 40,000 + 1,000) / 40,000 = 12.50%
Year 2 return = (46,000 − 44,000 + 1,000) / 44,000 = 6.82% (per share
base is the t = 1 price)
TWR = $[(1.1250)(1.0682)]^{1/2} - 1 = 9.62\%$
BA II Plus: 1.125 × 1.068182 = 1.201705
press the square-root key → 1.09623, then − 1 = 0.0962 = 9.62%

7. b
D = 100 × 0.08 / 4 = $2.00 per quarter
$R = (1 + 0.10/12)^3 - 1 = 2.5209\%$ per quarter
$P_{11} = D / R = 2.00 / 0.025209 = \$79.34$ (one quarter before the first dividend)
$P_0 = 79.34 / (1.025209)^{11} = \$60.33$
BA II Plus: 1 + 0.10 ÷ 12 = y^x 3 = 1.025209, so R = 2.5209%
BA II Plus: 11 N, 2.5209 I/Y, 0 PMT, −79.34 FV, CPT PV = 60.33

8. c
$g = (1.92 / 1.35)^{1/4} - 1 = 9.2048\%$ (four growth periods, not five)
$D_1 = 1.92 \times 1.092048 = \$2.0967$
$P_0 = D_1 / (R - g) = 2.0967 / (0.13 - 0.092048) = \$55.25$
BA II Plus: 4 N, −1.35 PV, 0 PMT, 1.92 FV, CPT I/Y = 9.2048
BA II Plus: 1.92 × 1.092048 ÷ 0.037952 = 55.25

9. a
$D_4 = 1.80 \times 1.20 = 2.160$
$D_5 = 2.160 \times 1.20 = 2.592$
$D_6 = 2.592 \times 1.20 = 3.1104$
$P_6 = D_7 / (R - g) = 3.1104 \times 1.05 / (0.14 - 0.05) = \$36.288$ (built from D_7, added to D_6 in year 6)
$P_0 = 1.80/(1.14)^3 + 2.160/(1.14)^4 + 2.592/(1.14)^5 + (3.1104 + 36.288)/(1.14)^6 = \$21.79$
BA II Plus: CF, CF0 = 0, C01 = 0, F01 = 2, C02 = 1.80, C03 = 2.16, C04 = 2.592, C05 = 39.3984, then NPV, I = 14, CPT = 21.79

10. d
Stage 1 = $[2.50 \times 1.18 / (0.14 - 0.18)] \times [1 - (1.18/1.14)^6]$ = (−73.75)(−0.22988) = $16.95 (both factors are negative, so the result is positive)
$D_6 = 2.50 \times (1.18)^6 = 6.7489$
$P_6 = 6.7489 \times 1.05 / (0.14 - 0.05) = \$78.74$
$P_0 = 16.95 + 78.74 / (1.14)^6 = 16.95 + 35.87 = \$52.83$
BA II Plus: 1.14 ÷ 1.18 − 1 = −0.033898 (I/Y = −3.3898)
BA II Plus: 6 N, −3.3898 I/Y, −2.50 PMT, 0 FV, CPT PV = 16.95
BA II Plus: 6 N, 14 I/Y, 0 PMT, −78.74 FV, CPT PV = 35.87

11. d
$D_t = 1.00 \times (1.12)^t$, and g = R, so every dividend has PV = $1.00
PV of dividends = 5 × 1.00 = $5.00
$\mathbf{EPS_5} = 3.00 \times (1.10)^5 = 4.8315$
$\mathbf{EPS_6} = 4.8315 \times 1.10 = 5.3147$
$\mathbf{P_5}$ = 14 × 5.3147 = $74.41 (forward P/E is applied to year-6 EPS)
PV of $\mathbf{P_5} = 74.41 / (1.12)^5 = \$42.22$
$\mathbf{P_0}$ = 5.00 + 42.22 = $47.22
BA II Plus: CF, CF0 = 0, C01 = 1.12, C02 = 1.2544, C03 = 1.404928, C04 = 1.573519, C05 = 76.1679, then NPV, I = 12, CPT = 47.22